PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

FLUE GAS WASHING POST

烟气洗涤岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the flue gas washing post under complete closure.

此程序涵盖操作员对完全关闭下的烟气洗涤岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the exhaust absorption post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the exhaust absorption post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对尾气吸收岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定尾气吸收岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP岗位操作规程
- Cooling Tower(4306-TOW-001). 冷却塔(4306-TOW-001)
- Cooling Tower Circulating Tank(4306-TNK-004/005). 冷却塔循环槽(4306-TNK-004/005)
- Deep Cone Thickener(4306-THK-001/002). 深锥浓密机(4306-THK-001/002)
- Gold Mud Pump(4306-PSL-001/002/011/012). 金泥输送泵(4306-PSL-001/002/011/012)
- Cooling Tower Circulating Pump(4306-PSL-003/004/005/006/007/020). 冷却塔循环泵(4306-PSL-003/004/005/006/007/020)
- Reverse Jet Scrubber(4306-TOW-002). 逆喷洗涤器(4306-TOW-002)
- Reverse Jet Scrubber Circulating Tank(4306-TNK-006). 逆喷洗涤器循环槽(4306-TNK-006)
- Reverse Jet Scrubber Circulating Pump(4306-PSL-008/009/010). 逆喷洗涤器循环泵(4306-PSL-008/009/010)
- Packed Scrubber Tower(4306-TOW-003). 填料洗涤塔(4306-TOW-003)
- Packed Scrubber Tower Circulating Tank(4306-TNK-007). 填料洗涤塔循环槽(4306-TNK-007)
- Packed Scrubber Tower Circulating Pump(4306-PSL-013/014/015/016). 填料洗涤塔循环泵(4306-PSL-013/014/015/016)
- Indirect Cooler(4306-COO-001/002/003/004). 间冷器(4306-COO-001/002/003/004)
- Indirect Cooler(4306-COO-005/006). 间冷器(4306-COO-005/006)
- Indirect Cooler (4306-COO-005/006) Circulating Pump(4306-PSL-020/021/022). 间冷器

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(4306-COO-005/006)循环泵(4306-PSL-020/021/022)

- Electric Demister(4306-ESP-001/002). 电除雾器 (4306-ESP-001/002)
- Dilute Acid Tank(4306-TNK-008). 稀酸槽(4306-TNK-008)
- Dilute Acid pump(4306-PSO-001/002). 稀酸泵(4306-PSO-001/002)
- Chlorinated Solution Tank(4402-TNK-001). 氯化溶液储槽(4402-TNK-001)
- Chlorinated Solution Pump(4402-PSO-001/002). 氯化溶液泵(4402-PSO-001/002)

4. PPE 个人保护装备



5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Liquid corrosion; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Poisoning; 7. Object falling from high altitudes; 8. Liquid sputtering; 9. Noise. 1. 机械伤害; 2. 液体腐蚀; 3. 触电; 4. 湿滑的条件，绊倒危险; 5. 窒息; 6. 中毒;	1. Install a protective cover on the rotating part of the equipment; 2. Wear appropriate labor protection supplies, once the skin contact with hydrochloric acid solution, immediately rinse with a lot of water; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear good labor protection goods, Found smoke or liquid running phenomenon immediately treated and reported; 7. Wear labor protection goods and check the safety of stacking items on the operation platform; 8. Wear protective masks and gloves, as well as

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Description of Hazard / Environmental Issue 危害	Control Strategy 控制
7. 高空坠物； 8. 液体喷溅； 9. 噪音。	other protective products as required； 9. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 穿戴适宜的劳保用品，一旦皮肤接触到盐酸溶液，立即用大量清水冲洗； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清塔或维修作业执行审批程序； 6. 穿戴好劳保护品，发现烟气或液体跑冒现象立即处理并汇报； 7. 穿戴好劳保护品，检查各平台的物品堆放安全性； 8. 按要求戴好防护面具和防护手套，以及其它劳保护品； 9. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4306-TNK-003	Water Head Tank高位水槽
4306-TNK-009	Water Seal Tank水封槽
4306-PSU-001	Submerged Pump液下泵
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训； 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生；

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№	Stage	Key Points
		4. 与前后工序取得联系，并发出明确指令。
2	Inspect Cooling Tower(4306-TOW-001) 检查冷却塔(4306-TOW-001)	2. Remove all sundries in the cooling tower and other equipment to ensure cleanliness; 3. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 4. Check whether the acid resistant bricks and other corrosion resistant bricks in the tower are in good condition; 5. Check whether the liquid pipelines and gas pipelines are in good condition; 6. Open the new water brake valve (100GA) and fill the cooling tower circulation tank (4306-TNK-004 / 005) with water; 7. Open the cooling tower circulation pump (4306-PSL-003 / 004 / 005 / 006 / 007 / 020), and test whether the temperature of the outer pipe wall of the upper and lower coil is consistent; 8. Check whether the cooling tower circulation liquid inlet, outlet, overflow port, gas inlet and outlet, manholes are normal, whether the connection bolts are fastened, whether leakage or leakage; check whether the tower body leakage or leakage; 9. Ensure that the cooling tower drain pipe is immersed in the liquid level of deep cone machine (4306-THK-001) below 1.2~1.5m; 10. Open the butterfly valve (200BF) and add the water level of the high level tank (4306-TNK-003) to the overflow port. 1. 清除冷却塔等设备内的一切杂物，确保干净； 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常； 3. 检查塔体内耐酸砖等耐蚀砖是否完好； 4. 检查各液体管线、气体管路是否完好； 5. 开启新水闸阀(100GA)，将冷却塔循环槽(4306-TNK-004/005)注满水； 6. 开启冷却塔循环泵(4306-PSL-003/004/005/006/007/020)，检测上下盘管外管壁温度是否一致； 7. 检查冷却塔循环液进口、循环液出口、溢流口、气体进出口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 8. 保证冷却塔排液管浸没在深锥浓密机(4306-THK-001)

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№	Stage	Key Points
		液位1.2～1.5m以下; 9. 将蝶阀(200BF)开启，将高位水槽 (4306-TNK-003)水位加至溢流口。
3	Inspect Cooling Tower Circulating Tank(4306-TNK-004/005),Reverse Jet Scrubber Circulating Tank(4306-TNK-006),Packed Scrubber Tower Circulating Tank(4306-TNK-007),Dilute Acid Tank(4306-TNK-008),Chlorinated Solution Tank (4402-TNK-001) 检查冷却塔循环槽(4306-TNK-004/005)， 逆喷洗涤器循环槽(4306-TNK-006)， 填料洗涤塔循环槽(4306-TNK-007)，稀酸槽(4306-TNK-008)， 氯化溶液储槽(4402-TNK-001)，	1. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the storage tank and dilute acid tank leaks liquid; 2. Check the level control assembly is normal. 1. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，循环槽和稀酸槽是否漏液; 2. 检查液位控制组件是否正常。
4	Inspect Deep Cone Thickener (4306-THK-001/002) 检查深锥浓密机(4306-THK-001/002)	2. Check whether the slurry distributor of deep cone thickening machine, overflow weir, deep cone cover plate and cover plate bridge are in good condition; 3. Check whether the overflow pipeline is in good condition; 4. Check whether the connecting bolts are fastened; 5. Rinse the deep cone thickening machine with new water to remove debris, and discharge the sewage from the bottom outlet of the deep cone thickening machine to the pit on the ground; 6. Fill the machine with new water and check for leakage; 7. Check whether the discharge outlet of the machine, the discharge pipe (4306-TOW-001) and the discharge pipe of the washing tower (4306-TOW-002) are normal; 8. Ensure that the drainage pipe of the cooling tower (4306-TOW-001) is immersed in the liquid level of the deep cone machine (4306-THK-001) to meet the liquid sealing

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№	Stage	Key Points
		effect of the washing tower; 9. Ensure that the discharge pipe of the reverse spray washing tower (4306-TOW-002) is immersed in the deep cone thickening machine (4306-THK-002) liquid level below 1.2~1.5m, and meet the liquid sealing effect of the reverse spray washing tower; 10. Ensure that the discharge pipe of the packing tower (4306-TOW-003) is immersed in the liquid level of the deep cone thickener (4306-THK-002) below 1.2~1.5m, and meet the liquid sealing effect of the reverse spray washing tower. 1. 检查深锥浓密机矿浆分配器、溢流堰、深锥盖板、盖板过桥是否完好； 2. 检查溢流管路是否完好； 3. 检查各处连接螺栓是否紧固； 4. 用新水冲洗深锥浓密机，清除杂物，污水从深锥浓密机底流放矿口排到地面的地坑； 5. 往深锥浓密机注满新水，检查有无泄露； 6. 检查深锥浓密机底流放矿口、冷却塔(4306-TOW-001)排液管、逆喷洗涤塔(4306-TOW-002)排液管是否正常； 7. 保证冷却塔(4306-TOW-001)排液管浸没在深锥浓密机(4306-THK-001)液位1.2 ~ 1.5m以下，满足洗涤塔液封效果； 8. 保证逆喷洗涤塔(4306-TOW-002)排液管浸没在深锥浓密机(4306-THK-002)液位1.2 ~ 1.5m以下，满足逆喷洗涤塔液封效果； 9. 保证填料塔(4306-TOW-003)排液管浸没在深锥浓密机(4306-THK-002)液位1.2 ~ 1.5m以下，满足逆喷洗涤塔液封效果。
5	Inspect Gold Mud Pump (4306-PSL-001/002/011/012), Cooling Tower Circulating Pump (4306-PSL-003/004/005/006/007/020), Reverse Jet Scrubber Circulating	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal;

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№	Stage	Key Points
	Pump(4306-PSL-008/009/010), Packed Scrubber Tower Circulating Pump (4306-PSL-013/014/015/016), Dilute Acid pump(4306-PSO-001/002), Chlorinated Solution Pump(4402-PSO-001/002),Indirect Cooler Circulating Pump(4306-PSL-020/021/022) 检查金泥输送泵(4306-PSL-001/002/011/012)，冷却塔循环泵(4306-PSL-003/004/005/006/007/020)，逆喷洗涤器循环泵(4306-PSL-008/009/010)，填料洗涤塔循环泵(4306-PSL-013/014/015/016)，稀酸泵(4306-PSO-001/002)，氯化溶液泵(4402-PSO-001/002)，间冷器循环泵(4306-PSL-020/021/022)	5. Point to move to confirm the correct steering; 6. Check whether the voltage meter, signal lamp and other meters of the pump control cabinet are normal; 7. Check whether the pump unit has air, if so, should be excluded; 8. Gold Mud delivery pump (4306-PSL-001/002/011/012) a. Check whether the pump head is under the liquid level of the deep cone machine (4306-THK-001/02); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (500BF) is temporarily closed after flexible flexibly; c. Check the outlet butterfly valve of the backup pump (400BF). 9. Cooling Tower circulating pump (4306-PSL-003/004/005/006/007/020) a. Check whether the pump head is under the liquid level of the cooling tower circulation tank (4306-TNK-004/005); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (125BF) is temporarily closed after flexible flexibly; c. Check the backup pump outlet butterfly valve (125BF) full clearance. 10. Circulation pump (4306-PSL-008/009/010) a. Check that the pump head is at the level of the sink (4306-TNK-006); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (500BF) is temporarily closed after flexible flexibly; c. Check that the backup pump outlet butterfly valve (500BF) is fully closed. 11. Filler washer pump (4306-PSL-013/014/015/016) a. Check that the pump head is at the level of the fill-washer circulation tank (4306-TNK-007); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (500BF) is temporarily closed after flexible flexibly; c. Check that the backup pump outlet butterfly valve (500BF) is fully closed. 12. Thin acid pump (4306-PSO-001/002) a. Check whether the pump head is under the level of dilute acid tank (4306-TNK-008); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (150BF) is temporarily closed after flexible flexibly; c. Check that the backup pump outlet butterfly valve

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		(150BF) is fully closed. 13. Check chlorinated solution pump(4402-PSO-001/002) a. Check whether the pump head is under the level of Chlorinated Solution Tank (4402-TNK-001); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (200BF) is temporarily closed after flexible flexibly; c. Check that the backup pump outlet butterfly valve (200BF) is fully closed. 14. Intercooler circulating pump (4306-PSL-020/021/022) a. Check whether the pump head is at the liquid level of the scrubber circulation tank (4306-TNK-007); b. Check that the delivery pump is in normal state, and the outlet butterfly valve (600BF) is temporarily closed after flexible flexibly; c. Check the backup pump outlet butterfly valve (600BF) full clearance. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好； 2. 手转动泵轴3圈，应灵活无阻滞； 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3之间； 4. 检查泵的密封状况是否正常； 5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 金泥输送泵(4306-PSL-001/002/011/012) a. 检查泵头是否处于深锥浓密机(4306-THK-001/002)液位下； b. 检查输送泵处于正常状态，出口蝶阀(500BF)开启灵活后暂时处于关闭状态； c. 检查备用泵出口蝶阀(400BF)关闭状态。 9. 冷却塔循环泵(4306-PSL-003/004/005/006/007/020) a. 检查泵头是否处于冷却塔循环槽(4306-TNK-004/005)液位下； b. 检查输送泵处于正常状态，出口蝶阀(125BF)开启灵活后暂时处于关闭状态； c. 检查备用泵出口蝶阀(125BF)关闭状态。

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№	Stage	Key Points
		10. 逆喷洗涤器循环泵(4306-PSL-008/009/010) a. 检查泵头是否处于逆喷洗涤器循环槽(4306-TNK-006)液位下; b. 检查输送泵处于正常状态，出口蝶阀(500BF)开启灵活后暂时处于关闭状态; c. 检查备用泵出口蝶阀(500BF)关闭状态。 11. 填料洗涤塔循环泵(4306-PSL-013/014/015/016) a. 检查泵头是否处于填料洗涤塔循环槽(4306-TNK-007)液位下; b. 检查输送泵处于正常状态，出口蝶阀(500BF)开启灵活后暂时处于关闭状态; c. 检查备用泵出口蝶阀(500BF)关闭状态。 12. 稀酸泵(4306-PSO-001/002) a. 检查泵头是否处于稀酸槽(4306-TNK-008)液位下; b. 检查输送泵处于正常状态，出口蝶阀(150BF)开启灵活后暂时处于关闭状态; c. 检查备用泵出口蝶阀(150BF)关闭状态。 13. 检查氯化溶液泵(4402-PSO-001/002) a. 检查泵头是否处于氯化溶液储槽(4402-TNK-001)液位下; b. 检查输送泵处于正常状态，出口蝶阀(200BF)开启灵活后暂时处于关闭状态; c. 检查备用泵出口蝶阀(200BF)关闭状态。 14. 间冷器循环泵(4306-PSL-020/021/022) a. 检查泵头是否处于洗涤塔循环槽(4306-TNK-007)液位下; b. 检查输送泵处于正常状态，出口蝶阀(600BF)开启灵活后暂时处于关闭状态; c. 检查备用泵出口蝶阀(600BF)关闭状态。
6	Inspect Reverse Jet Scrubber(4306-TOW-002) 检查逆喷洗涤器(4306-TOW-002)	1. Remove all sundries in the spray tower and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the new water brake valve (100GA) and fill the reverse spray washer circulation tank (4306-TNK-006) with water; 5. Open the circulating pump of the reverse spray washer

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№	Stage	Key Points
		(4306-PSL-008/009/010), observe the spray washing situation, whether the washing amount and washing density meet the requirements; 6. Check whether the overflow weir and nozzle components are normal and have no blockage phenomenon; 7. Check whether the circulating liquid inlet, circulating liquid outlet, backup port, gas inlet and outlet, manhole, and hand hole are normal, whether the connection bolts are fastened, whether leakage or leakage; check whether the tower body leakage or leakage; 8. Empty the inspection water in the reverse jet scrubber; 9. Ensure that the discharge pipe of the reverse spray tower is submerged below the liquid level of the deep cone thickener (4306-THK-002), and meet the liquid sealing effect of the reverse spray tower. 1. 清除逆喷洗涤塔等设备内的一切杂物; 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常; 3. 检查各液体管线、气体管路、阀门是否完好; 4. 开启新水闸阀(100GA)，将逆喷洗涤器循环槽(4306-TNK-006)注满水; 5. 开启逆喷洗涤器循环泵(4306-PSL-008/009/010)，观察喷淋洗涤情况，喷淋量、喷淋密度是否符合要求; 6. 检查逆喷洗涤塔溢流堰、喷嘴组件是否正常，有无堵塞现象; 7. 检查逆喷洗涤塔循环液进口、循环液出口、备用口、气体进出口、人孔、手孔等是否正常，连接螺栓是否紧固，是否漏液或漏气;检查塔体是否漏液或漏气; 8. 放空逆喷洗涤塔内的检查用水; 9. 保证逆喷洗涤塔排液管浸没在深锥浓密机(4306-THK-002)液位1.2～1.5m以下，满足逆喷洗涤塔液封效果。
7	Inspect Packed Scrubber Tower(4306-TOW-003) 检查填料洗涤塔(4306-TOW-003)	1. Remove all sundries in the packing material washing tower and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the front butterfly valve (100BF) of the filler washer tank, fill the circulation tank (4306-TNK-007) with the pumped washing water (4402-PSO-014~018), open the intercooler (4306-COO-005/006) butterfly

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		valve (600BF), open the filler washer spray butterfly valve (600BF), and open the packing washer circulation pump (4306-PSL-013/014/015/016); 5. Observe the spray washing situation of the packing washer, whether the spray amount and spray density meet the requirements; 6. Check whether the spray support ring, fog gate plate, screen separator and nozzle assembly are normal, and the spray remover and nozzle have no blockage; 7. Check whether the packing, compactness and height of the packing meet the requirements, and whether the spray dispersion is uniform; 8. Check whether the circulating fluid inlet, sewage outlet, backup outlet, gas inlet and outlet, manhole are normal, whether the connection bolts are fastened, whether leakage or leakage; check whether the tower body leakage or leakage; 9. Empty the inspection water in the packed scrubber tower; 10. Ensure that the discharge pipe of the packing washer is immersed below the liquid level 1.2~1.5m in the circulation tank (4306-TNK-007) to meet the liquid sealing effect of the packing washer. 1. 清除填料洗涤塔等设备内的一切杂物； 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常； 3. 检查各液体管线、气体管路、阀门是否完好； 4. 打开填料洗涤塔循环槽(4306-TNK-007)前蝶阀(100BF)，将循环槽注满循环泵(4402-PSO-014 ~ 018)泵过来的洗涤水，打开间冷器(4306-COO-005/006)进出蝶阀(600BF)，打开填料洗涤塔喷淋蝶阀(600BF)，开启填料洗涤塔循环泵(4306-PSL-013/014/015/016)； 5. 观察填料洗涤塔喷淋洗涤情况，喷淋量、喷淋密度是否符合要求； 6. 检查喷淋支座环、除雾栅板、丝网除沫器、喷嘴组件是否正常，除沫器和喷嘴有无堵塞现象； 7. 检查填料排布、密实度和高度是否符合要求，喷淋液散布是否均匀； 8. 检查循环液进口、排污口、备用口、气体进出口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 9. 放空填料洗涤塔内的检查用水；

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		10. 保证填料洗涤塔排液管浸没在循环槽(4306-TNK-007)液位1.2～1.5m以下，满足填料洗涤塔液封效果。
8	Inspect Indirect Cooler(4306-COO-001/002/003/004),(4306-COO-005/006) 检查间冷器(4306-COO-001/002/003/004)，(4306-COO-005/006)	1. Remove all sundries in the room cooler and other equipment; 2. Check whether the instruments are complete, and whether the pressure and thermocouple display is normal; 3. Check whether the liquid pipeline, gas pipeline and valves are in good condition; 4. Open the front gate valve of the circulation groove of the intercooler (400GA), fill the shell process of the intercooler with new water, open the two gate valves (4306-COO-001 / 002 / 003 / 004) (4306-COO-005 / 006) (300GA,300GA), check the shell process of the flushing room cooler, wash the water flow to the cooling circulating water station in the AP area;; 5. Check whether the inlet and outlet of circulating liquid, gas inlet and outlet, manhole, etc. are normal, whether the connection bolts are fastened, whether the leakage or leakage, and the pressure of pipe and shell process is normal; 6. Intercooler (4306-COO-001/002/003/004) (4306-COO-005/006)condensing bottom valve is flexible. 1. 清除间冷器等设备内的一切杂物； 2. 检查各处仪表是否齐全，压力和热电偶显示是否正常； 3. 检查各液体管线、气体管路、阀门是否完好； 4. 打开间冷器循环槽前闸阀(400GA)，将间冷器壳程注满新水，打开间冷器(4306-COO-001/002/003/004)壳程出口闸阀(400GA)、(4306-COO-005/006)两个闸阀(300GA,300GA)，检查冲洗间冷器的壳程，冲洗水流往AP区的冷却循环水站； 5. 检查循环液进出口、气体进出口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气，管程、壳程压力正常； 6. 间冷器(4306-COO-001/002/003/004)(4306-COO-005/006)冷凝水底阀是否灵活。
9	Inspect Electric Demister(4306-ESP-001/002) 检查电除雾器 (4306-ESP-001/002)	1. Check that the trachea, water pipe and valve are intact and reliable. The valve is open and closed at the specified position, and each manhole is closed. 2. Check whether the instruments are complete and the pressure and thermocouple display are normal; the constant temperature detection instrument is sensitive

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		and reliable and can automatically adjust the temperature to the specified value; 3. Check that the primary voltage and primary current of the power supply system are normal; 4. Check and adjust the temperature of high voltage rectifier unit, insulation box and the installation position of sensor are accurate; 5. Ensure whether the safety water seal of the pipe at the bottom of the electric fog ister is normal; 6. Ensure that the electric mist removal washing pump and other equipment and pipelines are normal; 7. Before starting the car, open the gate valve and ball valve (150GA, 150BA), spray the electric field, and drain the flushing water into the dilute acid tank (4306-TNK-008). 1. 检查气管、水管道、阀门完好可靠，阀门开、闭在规定位置，各人孔关闭。 2. 检查各处仪表是否齐全，压力和热电偶显示是否正常；恒温检测仪表灵敏可靠，能自动调节温度到规定值； 3. 检查供电系统的一次电压、一次电流正常； 4. 检查调整高压整流机组、绝缘箱温度、传感器安装位置是否准确； 5. 确保电除雾器下风头底部管路的安全水封是否正常； 6. 确保电除雾冲洗泵等设备、管路正常； 7. 开车前打开电除雾器上方闸阀和球阀(150GA,150BA)喷淋水阀，进行电场内部喷淋冲洗，冲洗水排往稀酸槽(4306-TNK-008)。
10	Check the other Equipment and Facilities 检查其它设备设施	1. Water head tank(4306-TNK-003) and water sealing tank (4306-TNK-009) a. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the high water tank and water sealing tank leaks liquid; b. Check the level control assembly is normal. 1. 高位水槽 (4306-TNK-003)和水封槽 (4306-TNK-009) a. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，高位水槽和水封槽是否漏液； b. 检查液位控制组件是否正常。 2. Submerged pump(4306-PSU-001) a. Check whether there is debris in the lower pump tank and the pipeline blockage, Whether the outlet

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		butterfly valve (65BF) is normal; b. Check whether the fastening screws of the equipment and motor are loose, whether the bearing is intact, and whether the lubricating oil is intact; c. Check whether the electrical part of the equipment is in good condition and whether the motor shell is well grounded; d. Check whether the steering of the motor is correct; e. Check whether the pump head is under the ground trough liquid level. 2. 液下泵(4306-PSU-001) a. 检查液下泵槽内有无杂物，管路堵塞情况，出口蝶阀(65BF)是否正常； b. 检查设备、电机的紧固螺丝是否松动，轴承是否完好，润滑油是否完好； c. 检查设备电气部分是否完好，电机外壳是否接地良好； d. 检查电机的转向是否正确； e. 检查泵头是否处于地槽液位下。 3. Valve a. Area valve of the cooling tower (4306-TOW-001) i. Check the two butterfly valves at the inlet of the high level tank (4306-TNK-003) (200BF, 200BF), dilute acid, two butterfly valves (150BF, 150BF), a butterfly valve at the outlet (200BF); High level sink (4306-TNK-003) at the entrance of tower 1 and tower 2 (4306-TOW-001) Butterfly valve (200BF, 200BF); Cooling Tower (4306-TOW-001) Tower 1 from cooling tower circulation tank (4306-TNK-004) Two butterfly valves (500BF, 500BF), tower No.2 from the cooling tower circulation tank (4306-TNK-005) (500BF, 500BF, 500BF, 500BF, 500BF, 500BF); Cooling tower (4306-TOW-001) Tower 1, Tower 2, bottom butterfly valve (800BF, 800BF) is intact and flexible; ii. Check the new water gate valve (100GA), the gold mud filter pump (4402-PSL-011) (4306-TNK-004) and a butterfly valve (80BF); The circulation groove (4306-TNK-004) and the circulation groove (4306-TNK-005) are connected to the butterfly valve (400BF); A drain butterfly valve (100BF) at

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		the bottom of the circulation tank (4306-TNK-004); Cooling tower circulating pump (4306-PSL-003/004/020) inlet four butterfly valves (400BF, 400BF, 400BF, 400BF), outlet, three butterfly valves (500BF, 500BF, 500BF) is intact and flexible; iii. Check the outlet butterfly valve of lower pump (4306-PSU-001) (65BF), deep cone machine (4306-THK-001) inlet butterfly valve (65BF); Deep cone machine (4306-THK-001) bottom flow two butterfly valves (200BF, 200BF); Deep cone machine (4306-THK-001) overflow to the chlorination solution tank (4402-TNK-001) a butterfly valve (150BF); Pump (4306-PSL-001/002) (200BF, 200BF), the outlet of the five butterfly valves (125BF, 125BF, 125BF, 125BF, 125BF) is intact and flexible; iv. Check the reverse spray washer circulating pump (4306-PSL-008/009/010) for a butterfly valve (110BF), deep cone thickener (4306-THK-001) and a butterfly valve (1200BF); The circulation groove (4306-TNK-005) and the circulation groove (4306-TNK-004) connect the butterfly valve (400BF); A drain butterfly valve (100BF) at the bottom of the circulation tank (4306-TNK-005); Cooling tower circulating pump (4306-PSL-005/006/007) inlet four butterfly valves (400BF, 400BF, 400BF, 400BF), outlet, three butterfly valves (500BF, 500BF, 500BF) is intact and flexible. b. Inwasher (4306-TOW-002) area valves i. Check a new water brake valve (100GA), a new water butterfly valve (80BF) in the reverse spray washer circulation tank (4306-TNK-006), A butterfly valve (100BF) pumped by the iron removal post washing pump (4402-PSO-017); A drain butterfly valve (100BF) at the bottom of the circulation tank (4306-TNK-006); Circulation pump (4306-PSL-008/009/010) inlet four butterfly valves (400BF, 400BF, 400BF, 400BF), outlet, three butterfly valves (500BF, 500BF, 500BF) is intact and flexible; ii. Check whether a butterfly valve (500BF) and a butterfly valve (500 BF) of the (4306-TOW-002) to the deep cone thickener (4306-THK-002) are intact and flexible; iii. Check the mud of one butterfly valve (65BF) pumped by the lower pump (4306-PSU-001)

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		(4306-THK-002);Deep cone machine (4306-THK-002) bottom flow two butterfly valves (200BF, 200BF);Whether the two butterfly valves (200BF, 200BF) and (4306-PSL-011/012) at the outlet (125BF, 125BF) are intact and flexible; c. Washer (4306-TOW-003) area valves i. Check the dilute acid valve (150BF) from the washing tank (4306-TNK-007) (4306-PSO-001/002), a new water gate valve (80GA), the pump (4402-PSO-017), a butterfly valve (100BF), a gate valve (80GA) (4306-TNK-007); Filler washer circulation pump (4306-PSL-013/014/015/016) inlet four butterfly valves (400BF, 400BF, 400BF, 400BF) and outlet five butterfly valves (600BF, 600BF, 600BF, 600BF, 100BF) are intact and flexible; ii. Check a butterfly valve (600BF) from the inter cooler (4306-COO-005/006) of packing washer (4306-TOW-003) for intact and flexible spray; iii. Check A new water brake valve (50GA) in the dilute acid tank (4306-TNK-008), A dilute acid one ball valve (150BF) from inter cooler (4306-COO-001/002/003/004), one ball valve (150BF) from electric mist (4306-ESP-001 / 002) and water seal (4306-TNK-009), A butterfly valve (100BF) at the bottom dilute acid discharge outlet; Sparute acid pump (4306-PSO-001/002) inlet three butterfly valves (200BF, 200BF, 200BF), outlet, two butterfly valves (150BF, 150BF). d. Inter cooler (4306-COO-001/002/003/004)(4306-COO-005 / 006) area valve i. A new water gate valve (400GA) (4306-COO-001/002), dilute acid pump (4306-PSO-001/002), filling sink (4306-TNK-007), inter cooler (4306-COO-001/002) inlet flue gas two butterfly valves (1500BF), shell outlet a gate valve (450GA), bottom discharge a ball valve (150BA) are intact and flexible; ii. Check whether a new water gate valve (400GA) of the inter cooler (4306-COO-003/004), a dilute acid pump (4306-PSO-001/002), an inter cooler (4306-COO-001/002), a butterfly valve (1300BF), a gate valve (400GA), a bottom discharge valve (150BA), and a butterfly valve (1200BF) are intact and flexible; iii. Check intercooler (4306-COO-005 / 006) two new

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		water brake valves (300GA, 300GA), whether the tube process calcium chloride solution into the four outlet butterfly valve (600BF,600BF,600BF,600BF), the two shell process cooling water outlet gate valve (300GA,300GA) and the two bottom ball valve (50BA,50BA) of the intercooler are intact and flexible; iv. Check whether the six butterfly valves (400BF,400BF,400BF,600BF, 600BF,600BF) circulating pump (4306-4306-005/006) (4306-PSL-020/021/022) are intact and flexible. e. Electric mist remover (4306-ESP-001/002) area valve i. Check the new water bulb valve (150BA), a ball valve (150BA) (4306-ESP-001/002) (150BA); a new water ball valve (4306-TNK-009). f. Valve in the storage tank of chlorination solution (4402-TNK-001) i. Check a new water brake valve (50GA) in the storage tank of chlorination solution (4402-TNK-001), a butterfly valve (150BF) from the deep cone machine (4306-THK-001), and whether a butterfly valve (100BF) at the bottom outlet is intact and flexible; ii. Check whether the chlorinated solution pump (4402-PSO-001/002), the inlet three butterfly valves (250BF, 250BF, 250BF) and the outlet two butterfly valves (200BF, 200BF) are intact and flexible. 3. 阀门 a. 冷却塔(4306-TOW-001)区域阀门 i. 检查高位水槽 (4306-TNK-003)入口的新水两个蝶阀(200BF,200BF)、稀酸两个蝶阀(150BF,150BF)、出口的一个蝶阀(200BF)；冷却塔(4306-TOW-001)1号塔、2号塔入口的高位水槽 (4306-TNK-003)蝶阀(200BF,200BF)；冷却塔(4306-TOW-001)1号塔来自冷却塔循环槽(4306-TNK-004)的喷淋液两个蝶阀(500BF,500BF)、2号塔来自冷却塔循环槽(4306-TNK-005)的喷淋液六个蝶阀

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		(500BF,500BF,500BF,500BF,500BF,500BF)；冷却塔(4306-TOW-001)1号塔、2号塔底部蝶阀(800BF,800BF)是否完好、灵活； ii. 检查循环槽(4306-TNK-004)的新水一个闸阀(100GA)、金泥滤液泵(4402-PSL-011)泵过来的滤液一个蝶阀(80BF)；循环槽(4306-TNK-004)与循环槽(4306-TNK-005)连通蝶阀(400BF)；循环槽(4306-TNK-004)底部一个放水蝶阀(100BF)；冷却塔循环泵(4306-PSL-003/004/020)入口四个蝶阀(400BF,400BF,400BF,400BF)、出口三个蝶阀(500BF,500BF,500BF)是否完好、灵活； iii. 检查液下泵(4306-PSU-001)出口蝶阀(65BF)、深锥浓密机(4306-THK-001)入口蝶阀(65BF)；深锥浓密机(4306-THK-001)底流两个蝶阀(200BF,200BF)；深锥浓密机(4306-THK-001)溢流去往氯化溶液储槽(4402-TNK-001)一个蝶阀(150BF)；金泥输送泵(4306-PSL-001/002)入口两个蝶阀(200BF,200BF)、出口五个蝶阀(125BF,125BF,125BF,125BF,125BF)是否完好、灵活； iv. 检查循环槽(4306-TNK-005)的逆喷洗涤器循环泵(4306-PSL-008/009/010)泵过来的吸收液一个蝶阀(110BF)、深锥浓密机(4306-THK-001)溢流液一个蝶阀(1200BF)；循环槽(4306-TNK-005)与循环槽(4306-TNK-004)连通蝶阀(400BF)；循环槽(4306-TNK-005)底部一个放水蝶阀(100BF)；冷却塔循环泵(4306-PSL-005/006/007)入口四个蝶阀(400BF,400BF,400BF,400BF)、出口三个蝶阀(500BF,500BF,500BF)是否完好、灵活。

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		b. 逆喷洗涤器(4306-TOW-002)区域阀门 i. 检查逆喷洗涤器循环槽(4306-TNK-006)的一个新水闸阀(100GA)、一个新水蝶阀(80BF)，除铁岗位洗涤水泵(4402-PSO-017)泵过来的洗水一个蝶阀(100BF)；循环槽(4306-TNK-006)底部一个放水蝶阀(100BF)；逆喷洗涤器循环泵(4306-PSL-008/009/010)入口四个蝶阀(400BF,400BF,400BF,400BF)、出口三个蝶阀(500BF,500BF,500BF)是否完好、灵活； ii. 检查逆喷洗涤器(4306-TOW-002)喷淋液一个蝶阀(500BF)、逆喷洗涤器(4306-TOW-002)底部喷淋液自流往深锥浓密机(4306-THK-002)的一个蝶阀(600BF)是否完好、灵活； iii. 检查深锥浓密机(4306-THK-002)的液下泵(4306-PSU-001)泵过来的泥浆一个蝶阀(65BF)；深锥浓密机(4306-THK-002)底流两个蝶阀(200BF,200BF)；金泥输送泵(4306-PSL-011/012)入口两个蝶阀(200BF,200BF)、出口两个蝶阀(125BF,125BF)是否完好、灵活； c. 填料洗涤塔(4306-TOW-003))区域阀门 i. 检查填料洗涤塔循环槽(4306-TNK-007)的稀酸泵(4306-PSO-001/002)泵过来的稀酸一个蝶阀(150BF)、一个新水闸阀(80GA)、除铁岗位洗涤水泵(4402-PSO-017)泵过来的洗水一个蝶阀(100BF)、PP区过来的浓密回水一个闸阀(80GA)、循环槽(4306-TNK-007)底部一个放水蝶阀(100BF)；填料洗涤塔循环泵(4306-PSL-013/014/015/016)入口四个蝶阀(400BF,400BF,400BF,400BF)、出口五个蝶

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		阀(600BF,600BF,600BF,600BF,100BF)是否完好、灵活； ii. 检查填料洗涤塔(4306-TOW-003)的来自间冷器(4306-COO-005/006)的喷淋液一个蝶阀(600BF)是否完好、灵活； iii. 检查稀酸槽(4306-TNK-008)的一个新水闸阀(50GA)，一个来自间冷器(4306-COO-001/002/003/004)的稀酸一个球阀(150BF)、来自电除雾器 (4306-ESP-001/002)和水封槽 (4306-TNK-009)的稀酸一个球阀(150BF)，底部稀酸排放口一个蝶阀(100BF)；稀酸泵(4306-PSO-001/002)入口三个蝶阀(200BF,200BF,200BF)、出口两个蝶阀(150BF,150BF)是否完好、灵活。 d. 间冷器(4306-COO-001/002/003/004)(4306-COO-005/006)区域阀门 i. 检查间冷器(4306-COO-001/002)的一个新水闸阀(400GA)、稀酸泵(4306-PSO-001/002)泵过来的稀酸一个球阀(175BA)、间冷器(4306-COO-001/002)入口烟气两个蝶阀(1500BF)、壳程出水一个闸阀(450GA)、底部放液一个球阀(150BA)是否完好、灵活； ii. 检查间冷器(4306-COO-003/004)的一个新水闸阀(400GA)、稀酸泵(4306-PSO-001/002)泵过来的稀酸一个球阀(175BA)、间冷器(4306-COO-001/002)过来的烟气一个蝶阀(1300BF)、壳程出水一个闸阀(400GA)、底部放液一个球阀(150BA)、烟气出口一个蝶阀(1200BF)是否完好、灵活； iii. 检查间冷器(4306-COO-005/006)两个新水闸阀(300GA,300GA)，壳程冷却水出口闸

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		阀两个(300GA,300GA)，管程氯化钙溶液入出口蝶阀四个(600BF,600BF,600BF,600BF)和间冷器底部球阀两个(50BA,50BA)是否完好、灵活； iv. 检查间冷器(4306-COO-005/006)循环泵(4306-PSL-020/021/022)前后六个蝶阀(400BF,400BF,400BF,600BF,600BF,600BF)是否完好、灵活。 e. 电除雾器 (4306-ESP-001/002)区域阀门 i. 检查电除雾器 (4306-ESP-001/002)新水闸阀(150GA)一个球阀(150BA)、底部一个球阀(150BA)；水封槽 (4306-TNK-009)的一个新水球阀(50BA)、稀酸排放口一个蝶阀(65BA)是否完好、灵活。 f. 氯化溶液储槽(4402-TNK-001)区域阀门 i. 检查氯化溶液储槽(4402-TNK-001)的一个新水闸阀(50GA)，深锥浓密机(4306-THK-001)溢流过来的一个蝶阀(150BF)，底部排放口一个蝶阀(100BF)是否完好、灵活； ii. 检查氯化溶液泵(4402-PSO-001/002)入口三个蝶阀(250BF,250BF,250BF)、出口两个蝶阀(200BF,200BF)是否完好、灵活。
11	Inspect Materials Level 检查物料料位	1. Check whether the high tank (4306-TNK-003), the bottom of cooling tower (4306-TOW-001), cooling tower circulation tank (4306-TNK-004/005), and deep cone machine (4306-THK-001) and chlorination solution tank(4402-TNK-001) are filled with new water; 2. Check whether the bottom of the reverse spray washer (4306-TOW-002), the reverse spray washer circulation (4306-TNK-006), the bottom of the filler washer (4306-TOW-003), the filler washer (4306-TNK-007), and the deep cone thickening machine (4306-THK-002) are filled with new water; 3. Check whether the intercooler (4306-COO-001/002/003/004)(4306-COO-005/006) shell process is filled with new water; 4. Check whether the dilute acid tank (4306-TNK-008) is

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		filled with new water; 5. Check whether the water sealing tank (4306-TNK-009) is filled with new water; 6. Check whether each liquid line accumulates fluid; 7. Check that the electric demister (4306-ESP-001/002) has been refueled. 1. 检查高位水槽 (4306-TNK-003)、冷却塔(4306-TOW-001)底部、冷却塔循环槽(4306-TNK-004/005)、深锥浓密机(4306-THK-001)和氯化溶液储槽(4402-TNK-001)是否充满新水； 2. 检查逆喷洗涤器(4306-TOW-002)底部、逆喷洗涤器循环槽(4306-TNK-006)、填料洗涤塔(4306-TOW-003)底部、填料洗涤塔循环槽(4306-TNK-007)、深锥浓密机(4306-THK-002)是否充满新水； 3. 检查间冷器(4306-COO-001/002/003/004)(4306-COO-005/006)壳程是否充满新水； 4. 检查稀酸槽(4306-TNK-008)是否充满新水； 5. 检查水封槽 (4306-TNK-009)是否充满新水； 6. 检查各液体管路是否积液； 7. 检查电除雾器 (4306-ESP-001/002)高压发生器已加油。
12	Start-Up Of The Flue Gas Washing Post 烟气洗涤岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the flue gas. In order to ensure the protection of the environment, all the equipment needs to be opened later; meanwhile, confirm that the process related to the exhaust gas absorption post has been started and that the feeding conditions are available: Open the cooling tower (4306-TOW-001) ⟹ Start the cooling tower (4306-TOW-001) Open two butterfly valves (200BF, 200BF) at the inlet of the high level sink (4306-TNK-003) and fill them with new water ⟹ Open the new water brake valve (100GA) of the cooling tower circulation tank (4306-TNK-004) with new water; open the circulation tank (4306-400BF) (4306-TNK-400BF) and fill the cooling tower circulation tank (4306-TNK-005) with new water ⟹ Open two butterfly valves (500BF, 500BF) of cooling tower (4306-TOW-001) from cooling tower circulation tank (4306-TNK-004) ⟹ Open the cooling tower circulating pump (4306-PSL-003/004/020) outlet three butterfly valves (500BF, 500BF, 500BF), start the circulating pump (4306-PSL-003/004/020) and open the inlet four butterfly valves (400BF, 400BF, 400BF, 400BF) ⟹ Open the cooling tower (4306-TOW-001) (500BF, 500BF, 500BF, 500BF) (5306-TNK-005) ⟹

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		Open the cooling tower circulation pump (4306-PSL-005/006/007) outlet three butterfly valves (500BF, 500BF, 500BF), open the circulation pump (4306-PSL-005 006/007), open the inlet four butterfly valves (400BF, 400BF, 400BF, 400BF) ⟹ Start Deep aper Concentrator (4306-THK-001) Open the overflow butterfly valve (4306-THK-001), After the circulation tank (4306-TNK-005) is sufficient, close the deep cone concentrate (4306-THK-001) overflow butterfly valve (1200BF) and open the deep cone concentrate (4306-THK-001) to the chloride solution tank (4402-TNK-001) a butterfly valve (150BF) ⟹ After the chlorination solution storage tank (4402-TNK-001) is filled with overflow, Open the two butterfly valves (200BF, 200BF) (4402-PSO-001/002), Start the chlorinated solution pump (4402-PSO-001/002), Open the inlet of three butterfly valves (250BF, 250BF, 250BF), Delivery chlorinated solution to neutralization tank (4402-TNK-002/003) ⟹ When the bottom flow concentration of deep cone machine (4306-THK-001) reaches the requirement, Open the outlet of the delivery pump (4306-PSL-001/002) (125BF, 125BF, 125BF, 125BF), Start the mud pump (4306-PSL-001/002), Open the delivery pump (4306-PSL-001/002) inlet with two butterfly valves (200BF, 200BF), Open the (4306-THK-001) bottom flow two butterfly valves (200BF, 200BF), Pump the slurry slurry bottom flow to the slurry storage tank (4402-TNK-022) ⟹ Wait for notification, open the absorption liquid butterfly valve (110BF) from the reverse spray washer circulation pump (4306-PSL-008/009/010) of the circulation tank (4306-TNK-005) ⟹ Open the reverse washer (4306-TOW-002) Open the new water brake valve (100GA) and the new water butterfly valve (80BF) in the circulation tank of the reverse spray washer (4306-TNK-006) and fill them with new water close the new water brake valve (100GA) ⟹ Open the reverse spray washer (4306-TOW-002) spray liquid butterfly valve (500BF) ⟹ Open the outlet three butterfly valves (500BF, 500BF, 500BF) of the reverse washer pump (4306-PSL-008/009/010), start the reverse washer circulation pump (4306-PSL-008/009/010) and open the inlet four butterfly valves (400BF, 400BF, 400BF, 400BF) ⟹ Start deep cone machine (4306-THK-002) Open the butterfly valve (600BF) of the bottom spray of the reverse washer (4306-TOW-002) to the deep cone machine (4306-

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		THK-002) ⟹ When the bottom flow concentration of the deep cone machine (4306-THK-002) reaches the requirement, Open the gold mud delivery pump (4306-PSL-011/012) outlet of two butterfly valves (125BF, 125BF), Start the mud pump (4306-PSL-011/012), Open the delivery pump (4306-PSL-011/012) inlet with two butterfly valves (200BF, 200BF), Open (4306-THK-002) flow butterfly valve (200BF), Pump the bottom flow of the gold mud slurry to the gold mud slurry storage tank (4402-TNK-022) ⟹ Wait for notification, open a butterfly valve (100BF) from the washing pump (4402-PSO-017) ⟹ Open the packing washer tower (4306-TOW-003) Open the new water brake valve (80GA) of the packing washer recycling tank (4306-TNK-007) and fill it with new water ⟹ Open the spray butterfly valve (600BF) from the intercooler (4306-COO-005/006) of the packing washer (4306-TOW-003) ⟹ Open the packing washer circulation pump (4306-PSL-013/014/015/016) outlet four butterfly valves (600BF, 600BF, 600BF, 600BF), Start the packing washer cycle pump (4306-PSL-013/014/015/016), Open the packing washer circulation pump (4306-PSL-013/014/015/016) inlet four butterfly valves (400BF, 400BF, 400BF, 400BF) ⟹ Open the new water ball valve (50GA) of the dilute acid tank (4306-TNK-008) and fill with new water ⟹ Open the two butterfly valves of the dilute acid pump (4306-PSO-001/002) and the dilute acid butterfly valve (130BF) and 150BF) inlet (TOW-003), start the dilute acid pump (4306-PSO-001/002), and open the three butterfly valves (4306-PSO-001/002) (200BF,200BF,200BF) ⟹ Wait for notification, open the dilute acid ball valve from the intercooler (4306-COO-001/002/003/004) (150BF) ⟹ Wait for notification, open the dilute acid ball valve (150BF) from the electric demister (4306-ESP-001/002) and the water sealing tank (4306-TNK-009) ⟹ Waiting for notice, iron post washing pump (4402-PSO-017) pump washing water butterfly valve (100BF), PP area thick return gate valve (80GA) ⟹ Start Intercooler (4306-COO-005/006) Open the new water brake valve (300GA) of the intercooler (4306-COO-005/006), fill the shell of the intercooler with new water, open the shell outlet gate valve (300GA), and the shell overflows to the cooling tower of AP area ⟹ Open the intercooler (4306-COO-005/006) pipe process outlet butterfly valve of

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		calcium chloride solution (600BF) and calcium chloride solution inlet butterfly valve (600BF)⟹Open the intercooler circulating pump (4306-PSL-020/021/022) rear butterfly valve (600BF) and pump front butterfly valve (400BF) and start the intercooler circulating pump (4306-PSL-020/021/022)⟹ To stop, open the intercooler (4306-COO-005/006) to drain two ball valves (50BA, 50BA), and drain the solution to the fill tower recycling tank (4306-TNK-007)⟹Open the intercooler (4306-COO-001/002/003/004)Open the new water brake valve (400GA) of the intercooler (4306-COO-001/002), fill the shell of the intercooler with new water, close the new water brake valve (400GA), open the shell outlet gate valve (450GA), the shell overflow to the cooling tower of AP area⟹ Wait for notification, open the dilute acid ball valve (175BA) from the dilute acid pump (4306-PSO-001/002) and open intercooler (4306-COO-001 / 002) inlet flue gas two butterfly valves (1500BF)⟹Open the new water brake valve (400GA) of the intercooler (4306-COO-003/004), fill the shell of the intercooler with new water, close the new water brake valve (400GA), open the shell outlet gate valve (400GA), and the shell flow flows to the cooling tower of AP area ⟹②Open the intercooler (4306-COO-003/004) outlet two butterfly valves (1200BF) and inlet two butterfly valves (1300BF)⟹ Wait for notification, open the dilute acid ball valve (175BA) from the dilute acid pump (4306-PSO-001/002)⟹Open the electric fog eliminator (4306-ESP-001/002)Open the new water ball valve (50BA) in the water seal tank (4306-TNK-009) and fill it with the new water⟹ The electric heaters of the four heating boxes must be powered 24 hours in advance, and after 50℃, 10~15℃/h to 120~140℃⟹ After the completion of the electrical equipment inspection, the high voltage disconnecting switch is sent to the power transmission position. When the power is transmitted, first open the main power switch, and then press the self-test button. After normal, supply the power demister electric field for power supply and air test ⟹ After the electric field is normal, stop the power transmission, cut off the power supply of the rectifier transformer, high voltage output grounding, power failure heater, the air test time should not exceed 10 minutes ⟹ Switch the "manual / automatic" switch of the control cabinet of the high-voltage rectifier and the spray

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		electric valve control box to the "manual" side ⟹ Open the spray gate valve and ball valve (150GA, 150BA), spray the electric field, spray water into the precipitate electrode to wet the surface as much as possible. After 20 minutes, close the spray valve; open a ball valve (150BA) at the bottom of the spray water into the dilute acid tank (4306-TNK-008) ⟹ Press the power transmission button of the electric demister, and adjust the voltage to about 30kV first ⟹ Observe the change of the secondary voltage and current, slowly adjust the secondary voltage of the two-stage electric fog to control at about 45kV, and the pole line has no grounding discharge phenomenon ⟹ Switch the "manual / automatic" switch of the control cabinet of the high-voltage rectifier and the spray electric valve control box to the "automatic" side ⟹ The electric demister has been put into operation 1. 启动时应保证空载启动; 2. 开车顺序为沿烟气走向，为保护环境，从前往后开即顺向开车，各设备都需要延时开启；同时，确认尾气吸收岗位相关的工序已经开车并具备给料条件： 开启冷却塔(4306-TOW-001) 启动冷却塔(4306-TOW-001) 打开高位水槽 (4306-TNK-003)入口的新水两个蝶阀(200BF,200BF)，注满新水 ⟹ 打开冷却塔循环槽(4306-TNK-004)的新水闸阀(100GA)，注满新水；打开循环槽(4306-TNK-004)与冷却塔循环槽(4306-TNK-005)两个连通蝶阀(400BF,400BF)，将冷却塔循环槽(4306-TNK-005)注满新水 ⟹ 打开冷却塔(4306-TOW-001)1号塔来自冷却塔循环槽(4306-TNK-004)的喷淋液两个蝶阀(500BF,500BF) ⟹ 打开冷却塔循环泵(4306-PSL-003/004/020)出口三个蝶阀(500BF,500BF,500BF)、启动循环泵(4306-PSL-003/004/020)，打开入口四个蝶阀(400BF,400BF,400BF,400BF) ⟹ 打开冷却塔(4306-TOW-001)2号塔来自冷却塔循环槽(4306-TNK-005)的喷淋液四个直通蝶阀(500BF,500BF,500BF,500BF) ⟹ 打开冷却塔循环泵(4306-PSL-005/006/007)出口三个蝶阀(500BF,500BF,500BF)、开启循环泵(4306-PSL-005/006/007)，打开入口四个蝶阀(400BF,400BF,400BF,400BF) ⟹ 启动深锥浓密机(4306-THK-001) 打开深锥浓密机(4306-THK-001)溢流液蝶阀(1200BF)，待循环槽(4306-TNK-005)循环液足够后，关闭深锥浓密机(4306-THK-001)溢流液蝶阀

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		(1200BF)，打开深锥浓密机(4306-THK-001)溢流去往氯化溶液储槽(4402-TNK-001)一个蝶阀(150BF)⟹氯化溶液储槽(4402-TNK-001)充满溢流后，打开氯化溶液泵(4402-PSO-001/002)出口两个蝶阀(200BF,200BF)，启动氯化溶液泵(4402-PSO-001/002)，打开入口三个蝶阀(250BF,250BF,250BF)，将氯化溶液输送到中和槽(4402-TNK-002/003)⟹待深锥浓密机(4306-THK-001)底流浓度达到要求时，打开金泥输送泵(4306-PSL-001/002)出口四个直通蝶阀(125BF,125BF,125BF,125BF)，启动金泥输送泵(4306-PSL-001/002)，打开输送泵(4306-PSL-001/002)入口两个蝶阀(200BF,200BF)，打开深锥浓密机(4306-THK-001)底流上面一个蝶阀(200BF)，将金泥浆液底流泵往金泥储浆槽(4402-TNK-022)⟹等待通知，打开循环槽(4306-TNK-005)的逆喷洗涤器循环泵(4306-PSL-008/009/010)泵过来的吸收液蝶阀(110BF)⟹开启逆喷洗涤器(4306-TOW-002)启动逆喷洗涤器(4306-TOW-002)打开逆喷洗涤器循环槽(4306-TNK-006)的新水闸阀(100GA)、新水蝶阀(80BF)，注满新水后关闭新水闸阀(100GA)⟹打开逆喷洗涤器(4306-TOW-002)喷淋液蝶阀(500BF)⟹打开逆喷洗涤器循环泵(4306-PSL-008/009/010)的出口三个蝶阀(500BF,500BF,500BF)，启动逆喷洗涤器循环泵(4306-PSL-008/009/010)，打开入口四个蝶阀(400BF,400BF,400BF,400BF)⟹启动深锥浓密机(4306-THK-002)打开逆喷洗涤器(4306-TOW-002)底部喷淋液自流往深锥浓密机(4306-THK-002)的蝶阀(600BF)⟹待深锥浓密机(4306-THK-002)底流浓度达到要求时，打开金泥输送泵(4306-PSL-011/012)出口两个蝶阀(125BF,125BF)，启动金泥输送泵(4306-PSL-011/012)，打开输送泵(4306-PSL-011/012)入口两个蝶阀(200BF,200BF)，打开深锥浓密机(4306-THK-002)底流蝶阀(200BF)，将金泥浆液底流泵往金泥储浆槽(4402-TNK-022)⟹等待通知，打开除铁岗位洗涤水泵(4402-PSO-017)泵过来的洗水一个蝶阀(100BF)⟹开启填料洗涤塔(4306-TOW-003)启动填料洗涤塔(4306-TOW-003)打开填料洗涤塔循环槽(4306-TNK-007)的新水闸阀(80GA)，注满新水⟹打开填料洗涤塔(4306-TOW-003)的来自间冷器(4306-COO-

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		005/006)的喷淋液蝶阀(600BF)⟶打开填料洗涤塔循环泵(4306-PSL-013/014/015/016)出口四个蝶阀(600BF,600BF,600BF,600BF)，启动填料洗涤塔循环泵(4306-PSL-013/014/015/016)，打开填料洗涤塔循环泵(4306-PSL-013/014/015/016)入口四个蝶阀(400BF,400BF,400BF,400BF)⟶打开稀酸槽 (4306-TNK-008)的新水球阀(50GA)，注满新水⟶打开稀酸泵(4306-PSO-001/002)出口两个蝶阀(150BF,150BF)和填料洗涤塔(4306-TOW-003)入口稀酸蝶阀(100BF),启动稀酸泵(4306-PSO-001/002)，打开稀酸泵(4306-PSO-001/002)入口三个蝶阀(200BF,200BF,200BF)⟶等待通知，打开间冷器(4306-COO-001/002/003/004)过来的稀酸球阀(150BF)⟶等待通知，打开电除雾器(4306-ESP-001/002)和水封槽 (4306-TNK-009)过来的稀酸球阀(150BF)⟶等待通知，打开除铁岗位洗涤水泵(4402-PSO-017)泵过来的洗水蝶阀(100BF)、PP区过来的浓密回水闸阀(80GA)⟹启动间冷器(4306-COO-005/006)打开间冷器(4306-COO-005/006)的新水闸阀(300GA)，将间冷器壳程注满新水后，打开壳程出水闸阀(300GA)，壳程溢水流往AP区冷却塔⟶打开间冷器(4306-COO-005/006)管程氯化钙溶液出口蝶阀(600BF)和氯化钙溶液入口蝶阀(600BF)⟶打开间冷器循环泵(4306-PSL-020/021/022)后蝶阀(600BF)和泵前蝶阀(400BF)，启动间冷器循环泵(4306-PSL-020/021/022)⟶待停车时，打开间冷器(4306-COO-005/006)底部放液两个球阀(50BA,50BA)，放空溶液流往填料塔循环槽(4306-TNK-007)⟹开启间冷器(4306-COO-001/002/003/004)打开间冷器(4306-COO-001/002)的新水闸阀(400GA)，将间冷器壳程注满新水后，关闭新水闸阀(400GA)打开壳程出水闸阀(450GA)，壳程溢水流往AP区冷却塔⟶等待通知，打开稀酸泵(4306-PSO-001/002)泵过来的稀酸球阀(175BA)、打开间冷器(4306-COO-001/002)入口烟气两个蝶阀(1500BF)⟶打开间冷器(4306-COO-003/004)的新水闸阀(400GA)，将间冷器壳程注满新水后，关闭新水闸阀(400GA)打开壳程出水闸阀(400GA)，壳程溢水流往AP区冷却塔⟶打开间冷器(4306-COO-003/004)出口烟气两个蝶阀(1200BF)和入口烟气两个蝶阀(1300BF)⟶等待通知，打开稀酸泵(4306-PSO-001/002)泵过来的稀酸球阀(175BA)

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		⟹开启电除雾器(4306-ESP-001/002)｛打开水封槽(4306-TNK-009)的新水球阀(50BA)，注满新水⟹四个加热箱电加热器必须提前24小时送电，升温50℃后应按10～15℃/h，缓慢升温到120～140℃之间运行⟹电气设备检查工作完成后，高压隔离开关投到送电位置，送电时，先开启总电开关后，再按自检按钮，正常后，给电除雾器电场进行通电空试⟹电场一切正常后要停止送电，切断整流变压器电源,高压输出接地，停电加热器，空试时间不能超过10分钟⟹将高压整流机控制柜、喷淋电动阀控制箱的“手动/自动”开关切换至“手动”侧⟹打开喷淋水闸阀和球阀(150GA,150BA)，进行电场内部喷淋清洗，向沉淀电极管内喷水使其表面尽量湿润，20分钟后，关闭喷淋水阀；打开电除雾底部一个球阀(150BA)将冲洗水排到稀酸槽(4306-TNK-008)⟹按下电除雾器送电按钮，将电压先调到30kV左右⟹观察二次电压、电流的变化，缓慢将两级电雾的二次电压调至控制在45kV左右，极线无接地放电现象⟹将高压整流机控制柜、喷淋电动阀控制箱的“手动/自动”开关切换至“自动”侧⟹电除雾器投入运行操作完毕｝
13	The Flue Gas Washing Post Stops 烟气洗涤岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order , Each equipment needs to stop: Stop the electric defogging device (4306-ESP-001/002)｛ Short stop［Adjust the current and voltage adjustment knob to a minimum⟹Press the shutdown button⟹Cut off the power switch, disconnect the high voltage disconnecting switch, ground the high voltage disconnecting switch, stop the electric heater, hang the "No power transmission" sign on the corresponding operating plate and padlock it］⟹Emergency out-off［Press the "Emergency stop" button, the high voltage drops to 0 in an instant, and the electric fog eliminator stops running⟹Notify the relevant personnel after the electric fog stop device has stopped, cut off the power switch, turn off the high voltage disconnecting switch, the high voltage disconnecting switch grounding, stop the electric heater from running, hang the "No power transmission" sign on the corresponding operating

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		plate and padlock⟹Organize personnel to deal with the fault, after the processing, according to the manual driving operation⟹ Long term stop⟹ Adjust the current and voltage adjustment knob to the minimum, and press the stop button⟹Check that the low-voltage power supply is indeed power-free, put the high-voltage switching switch of the rectifier unit room into the grounding, open the door of the rectifier chamber, discharge the high-voltage part with the insulated discharge rod, and short-connect the high-voltage side with the grounding wire and ground⟹Hang the "someone works, not closed" sign on the control cabinet and padlock⟹ Open the manhole, open the spray gate valve and ball valve (150GA, 150BA), open a ball valve (150BA) at the bottom of the electric mist, flush the sedimentation tube, flush water into the dilute acid tank (4306-TNK-008); and discharge the water seal tank (4306-TNK-009) drain butterfly valve (65BA), open the water seal tank (4306-TNK-009) solution into the dilute acid tank (4306-TNK-008)⟹The electric mist remover is in a long-term parking state ⟹ Stop packing washer tower (4306-TOW-003)⟹Close the dilute acid pump (4306-PSO-001/002) inlet three butterfly valves (200BF,200BF,200BF), stop the dilute acid pump (4306-PSO-001/002), close the dilute acid pump (4306-PSO-001/002) outlet two butterfly valves (150BF,150BF) and filler washing tower (4306-TOW-003) inlet dilute acid butterfly valve (100BF)⟹ Close the packing washer circulation pump (4306-PSL-013/014/015/016) (400BF,400BF,400BF,400BF), stop the packing washer circulation pump (4306-PSL-013/014/015/016), and close the packing washer circulation pump (4306-PSL-013/014/015/016) (600BF,600BF,600BF,600BF)⟹ Spray butterfly valve (600BF) from the intercooler (4306-COO-005/006) closing the packing washer (4306-TOW-006)⟹ Close the dilute acid ball valve (150BF) from the intercooler (4306-COO-001/002/003/004) ⟹ Close the dilute acid ball valve (150BF) from the electric defister (4306-ESP-001/002) and the water sealing tank (4306-TNK-009)⟹ Close the washing water butterfly valve (100BF) from the washing pump (4402-PSO-017) and the dense return

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№	Stage	Key Points
		gate valve (80GA) from the PP area ⟹ Close the front pump butterfly valve (400BF), stop the intercooler circulation (4306-COO-005/006) pump (4306-PSL-020/021/022), and close the rear pump butterfly valve (600BF) ⟹ Close the new water gate valve (300BA) and the pipeline calcium chloride solution inlet butterfly valve (600BF), stop the intercooler (4306-COO-005/006), close the shell cooling water outlet gate valve (300GA) and the calcium chloride solution outlet butterfly valve (600BF) ⟹ For long-term parking, open the intercooler (4306-COO-005/006) to drain two ball valves (50BA, 50BA), and drain the vent solution to the packing tower recycling tank (4306-TNK-007) ⟹ Stop Room Cooler (4306-COO-001/002/003/004) Short-term stop and emergency stop, (1)Close the dilute acid ball valve (175BA) (4306-PSO-001/002), close the intercooler (4306-COO-001/002/003/004) shell outlet gate valve (450GA) ⟹ (2)Close the intercooler (4306-COO-001/002) inlet flue gas with two butterfly valves (1500BF) ⟹ (3)Two butterfly valves (1300BF) of inlet intercooler (4306-COO-003/004) and two butterfly valves (1200BF) ⟹ long term stopOpen the intercooler (4306-COO-001/002/003/004) to drain four ball valves (150BA, 150BA, 150BA, 150BA) into the dilute acid tank (4306-TNK-008) ⟹ Stop reverse washer (4306-TOW-002) Close the inlet circulating pump (4306-PSL-008/009/010) (400BF,400BF,400BF,400BF), stop the circulating pump (4306-PSL-008/009/010) and close the three butterfly valves (500BF,500BF,500BF) of the circulating pump (4306-PSL-008/009/010) ⟹ Close the reverse spray washer (4306-TOW-002) spray liquid butterfly valve (500BF) ⟹ After the discharge of the bottom flow of the reverse spray washer (4306-TOW-002), close the bottom spray washer (4306-TOW-002) to the butterfly valve (600BF) of the deep cone machine (4306-THK-002) ⟹ When the bottom flow concentration of deep cone dense machine (4306-THK-002) is lower than the emission requirement, close the deep cone machine (4306-THK-002) butterfly valve (200BF), close the two butterfly valves (200BF,200BF), stop

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№	Stage	Key Points
		the gold conveying pump (4306-PSL-011/012), and close the two butterfly valves (125BF,125BF) (4306-PSL-011/012)⟹ Close the washing water butterfly valve (100BF) pumped from the iron removal post washing water pump (4402-PSO-017)⟹ Stop cooling tower (4306-TOW-001)Stop cooling tower (4306-TOW-001)Close the cooling tower circulating pump (4306-PSL-005/006/007) inlet four butterfly valves (400BF, 400BF, 400BF, 400BF), stop the cooling tower circulating pump (4306-PSL-005/006/007), close the cooling tower circulating pump (4306-PSL-005/006/007) outlet three butterfly valves (500BF, 500BF, 500BF)⟹ Close the cooling tower (4306-TOW-001) (500BF, 500BF, 500BF, 500BF) (4306-TNK-005)⟹ Close the cooling tower circulating pump (4306-PSL-003/004/020) (400BF, 400BF, 400BF, 400BF), stop the cooling tower circulating pump (4306-PSL-003/004/020), close the cooling tower circulating pump (4306-PSL-003/004/020) (500BF, 500BF, 500BF)⟹ If stopped for a long time, open the cooling tower (4306-TOW-001) (800BF, 800BF, 800BF), empty all the solution to the deep cone thickener (4306-THK-001), and close the cooling tower (4306-TOW-001) (800BF, 800 BF)⟹ Stop Deep Cone Concentrator (4306-THK-001)When the bottom flow concentration of the deep cone machine (4306-THK-001) is below the emission requirement, Close the butterfly valve (200BF) above the bottom of the cone (4306-THK-001), Close the two butterfly valves (200BF, 200BF) (4306-PSL-001/002), Stop the gold mud delivery pump (4306-PSL-001/002), Close the outlet (426-PSL-001/002) (125BF, 125BF, 125BF, 125BF)⟹The absorption butterfly valve (110BF) pumped through the 4306-reverse spray washer circulating pump (4306-PSL-008/009/010) of the circulation tank (4306-TNK-005)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序顺序停车，各设备都需要延时停止： 停止电除雾器 (4306-ESP-001/002)短期停车将

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		电流和电压调节旋钮调到最小⟹ 按停机按钮⟹切断电源开关，断开高压隔离开关，高压隔离开关接地，电加热器停止运行，在相应的操作盘上挂上“严禁送电”标示牌并挂锁⟹紧急停车按下“紧急停车”按钮，高压电在瞬间降为0，电除雾器停止运行⟹ 通知相关人员电除雾器已紧急停车后，切断电源开关，断开高压隔离开关，高压隔离开关接地，电加热器停止运行，在相应的操作盘上挂上“严禁送电”标示牌并挂锁⟹组织人员处理故障，处理完毕后，按手动开车操作进行⟹长期停车将电流、电压调节旋钮调到最小，按下停机按钮⟹检查低压电源确实无电，将整流机组室高压切换开关打入接地，打开整流室门，用绝缘放电棒将高压部份放电，将高压侧用接地线短接并接地⟹控制柜上挂好“有人工作，不可合闸”标示牌并挂锁⟹折开人孔，打开喷淋水闸阀和球阀(150GA,150BA)，打开电除雾底部一个球阀(150BA)，将沉淀极管冲洗干净，冲洗水排到稀酸槽(4306-TNK-008)；并将水封槽(4306-TNK-009)排液口蝶阀(65BA)打开，水封槽(4306-TNK-009)溶液排到稀酸槽(4306-TNK-008)⟹电除雾器则处于长期停车状态⟹停止填料洗涤塔(4306-TOW-003)关闭稀酸泵(4306-PSO-001/002)入口三个蝶阀(200BF,200BF,200BF)，停止稀酸泵(4306-PSO-001/002)，关闭稀酸泵(4306-PSO-001/002)出口两个蝶阀(150BF,150BF)和填料洗涤塔(4306-TOW-003)入口稀酸蝶阀(100BF)⟹关闭填料洗涤塔循环泵(4306-PSL-013/014/015/016)入口四个蝶阀(400BF,400BF,400BF,400BF)，停止填料洗涤塔循环泵(4306-PSL-013/014/015/016)，关闭填料洗涤塔循环泵(4306-PSL-013/014/015/016)出口四个蝶阀(600BF,600BF,600BF,600BF)⟹关闭填料洗涤塔(4306-TOW-003)的来自间冷器(4306-COO-005/006)的喷淋液蝶阀(600BF)⟹关闭间冷器(4306-COO-001/002/003/004)过来的稀酸球阀(150BF)⟹关闭电除雾器 (4306-ESP-001/002)和水封槽 (4306-TNK-009)过来的稀酸球阀(150BF)⟹关闭除铁岗位洗涤水泵(4402-PSO-017)泵过来的洗水蝶阀(100BF)、PP区过来的浓密回水闸阀(80GA)⟹关闭间冷器(4306-COO-005/006)

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		循环泵(4306-PSL-020/021/022)前蝶阀(400BF)，停止间冷器循环泵(4306-PSL-020/021/022)，关闭间冷器循环泵(4306-PSL-020/021/022)后蝶阀(600BF)⟹关闭新水闸阀(300BA)和管程氯化钙溶液入口蝶阀(600BF)，停止间冷器(4306-COO-005/006)，关闭壳程冷却水出口闸阀(300GA)和氯化钙溶液出口蝶阀(600BF)⟹长期停车时，打开间冷器(4306-COO-005/006)底部放液两个球阀(50BA,50BA)，放空溶液流往填料塔循环槽(4306-TNK-007)⟹停止间冷器(4306-COO-001/002/003/004)短期停车和紧急停车(1)关闭稀酸泵(4306-PSO-001/002)泵过来的稀酸球阀(175BA)、关闭间冷器(4306-COO-001/002/003/004)壳程出水闸阀(450GA)⟹(2)关闭间冷器(4306-COO-001/002)入口烟气两个蝶阀(1500BF)⟹(3)关闭间冷器(4306-COO-003/004)入口烟气两个蝶阀(1300BF)和出口烟气两个蝶阀(1200BF)⟹长期停车打开间冷器(4306-COO-001/002/003/004)底部放液四个球阀(150BA,150BA,150BA,150BA)，放空稀酸流往稀酸槽(4306-TNK-008)⟹停止逆喷洗涤器(4306-TOW-002)关闭逆喷洗涤器循环泵(4306-PSL-008/009/010)入口四个蝶阀(400BF,400BF,400BF,400BF)，停止逆喷洗涤器循环泵(4306-PSL-008/009/010)，关闭逆喷洗涤器循环泵(4306-PSL-008/009/010)的出口三个蝶阀(500BF,500BF,500BF)⟹关闭逆喷洗涤器(4306-TOW-002)喷淋液蝶阀(500BF)⟹逆喷洗涤器(4306-TOW-002)底流排放结束后，关闭逆喷洗涤器(4306-TOW-002)底部喷淋液自流往深锥浓密机(4306-THK-002)的蝶阀(600BF)⟹待深锥浓密机(4306-THK-002)底流浓度低于排放要求时，关闭深锥浓密机(4306-THK-002)蝶阀(200BF)，关闭输送泵(4306-PSL-011/012)入口两个蝶阀(200BF,200BF)，停止金泥输送泵(4306-PSL-011/012)，关闭金泥输送泵(4306-PSL-011/012)出口两个蝶阀(125BF,125BF)⟹关闭除铁岗位洗涤水泵(4402-PSO-017)泵过来的洗水蝶阀(100BF)⟹停止冷却塔(4306-TOW-001)停止冷却塔

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		(4306-TOW-001)关闭冷却塔循环泵(4306-PSL-005/006/007)入口四个蝶阀(400BF,400BF,400BF,400BF)，停止冷却塔循环泵(4306-PSL-005/006/007)，关闭冷却塔循环泵(4306-PSL-005/006/007)出口三个蝶阀(500BF,500BF,500BF)⟹关闭冷却塔(4306-TOW-001)2号塔来自冷却塔循环槽(4306-TNK-005)的喷淋液四个直通蝶阀(500BF,500BF,500BF,500BF)⟹关闭冷却塔循环泵(4306-PSL-003/004/020)入口四个蝶阀(400BF,400BF,400BF,400BF)，停止冷却塔循环泵(4306-PSL-003/004/020)，关闭冷却塔循环泵(4306-PSL-003/004/020)出口三个蝶阀(500BF,500BF,500BF)⟹关闭冷却塔(4306-TOW-001)1号塔来自冷却塔循环槽(4306-TNK-004)的喷淋液两个蝶阀(500BF,500BF)⟹如果长时间停车，打开冷却塔(4306-TOW-001)1号塔、2号塔底部蝶阀(800BF,800BF)，将溶液全部排空至深锥浓密机(4306-THK-001)，然后关闭冷却塔(4306-TOW-001)1号塔、2号塔底部蝶阀(800BF,800BF)⟹停止深锥浓密机(4306-THK-001)待深锥浓密机(4306-THK-001)底流浓度低于排放要求时，关闭深锥浓密机(4306-THK-001)底流上面的那个蝶阀(200BF)，关闭输送泵(4306-PSL-001/002)入口两个蝶阀(200BF,200BF)，停止金泥输送泵(4306-PSL-001/002)，关闭金泥输送泵(4306-PSL-001/002)出口四个直通蝶阀(125BF,125BF,125BF,125BF)⟹关闭循环槽(4306-TNK-005)的逆喷洗涤器循环泵(4306-PSL-008/009/010)泵过来的吸收液蝶阀(110BF)⟹打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, Close cooling tower(4306-TOW-001), deep cone thickener(4306-THK-001/002), gold mud pump(4306-PSL-001/002/011/012), cooling tower circulating pump(4306-PSL-003/004/005/006/007/020), reverse jet scrubber(4306-TOW-002), reverse jet scrubber circulating pump(4306-PSL-008/009/010), packed scrubber tower(4306-TOW-003), packed scrubber tower circulating pump(4306-PSL-

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№	Stage	Key Points
		013/014/015/016), indirect cooler(4306-COO-001/002/003/004), electric demister(4306-ESP-001/002) and dilute acid pump(4306-PSO-001/002), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close cooling tower(4306-TOW-001), deep cone thickener(4306-THK-001/002), gold mud pump(4306-PSL-001/002/011/012), cooling tower circulating pump(4306-PSL-003/004/005/006/007/020), reverse jet scrubber(4306-TOW-002), reverse jet scrubber circulating pump(4306-PSL-008/009/010), packed scrubber tower(4306-TOW-003), packed scrubber tower circulating pump(4306-PSL-013/014/015/016), intercooler circulating pump (4306-PSL-020/021/022) and intercooler (4306-COO-005/006), indirect cooler(4306-COO-001/002/003/004), electric demister(4306-ESP-001/002), dilute acid pump(4306-PSO-001/002), close the pump front butterfly valve (400BF) and the pump rear butterfly valve (600BF) of the intercooler circulating pump (4306-PSL-020/021/022), New water brake valve (300BA) of closed intercooler (4306-COO-005/006), pipe calcium chloride solution inlet butterfly valve (600BF), butterfly valve of shell cooling water outlet valve (300GA), intercooler circulation (4306-COO-005/006) pump (4306-PSL-020/021/021/022), cooling tower circulating tank(4306-TNK-004/005), reverse jet scrubber circulating tank(4306-TNK-006), packed scrubber tower circulating tank(4306-TNK-007), water head tank(4306-TNK-003) and water seal tank(4306-TNK-009) corresponding valve, at the same time, the upper level of equipment all need to stop; In case of an emergency, first shut down the fan (4306-FAN-001/002/003), and then stop according to the parking order; b. After the accident, the equipment is return to normal. If the production is still needed, the operation shall be restarted in the order of driving; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order.

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№	Stage	Key Points
		2. 紧急停车 a. 当出现紧急事故，关闭冷却塔(4306-TOW-001)、深锥浓密机(4306-THK-001/002)、金泥输送泵(4306-PSL-001/002/011/012)、冷却塔循环泵(4306-PSL-003/004/005/006/007/020)、逆喷洗涤器(4306-TOW-002)、逆喷洗涤器循环泵(4306-PSL-008/009/010)、填料洗涤塔(4306-TOW-003)、填料洗涤塔循环泵(4306-PSL-013/014/015/016)、间冷器循环泵(4306-PSL-020/021/022)和间冷器(4306-COO-005/006)、间冷器(4306-COO-001/002/003/004)、电除雾器 (4306-ESP-001/002)和稀酸泵(4306-PSO-001/002)，间冷器(4306-COO-005/006)循环泵(4306-PSL-020/021/022)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭冷却塔(4306-TOW-001)、深锥浓密机(4306-THK-001/002)、金泥输送泵(4306-PSL-001/002/011/012)、冷却塔循环泵(4306-PSL-003/004/005/006/007/020)、逆喷洗涤器(4306-TOW-002)、逆喷洗涤器循环泵(4306-PSL-008/009/010)、填料洗涤塔(4306-TOW-003)、填料洗涤塔循环泵(4306-PSL-013/014/015/016)、关闭间冷器(4306-COO-005/006)循环泵(4306-PSL-020/021/022)的泵前蝶阀(400BF)和泵后蝶阀(600BF)，关闭间冷器(4306-COO-005/006)的新水闸阀(300BA)、管程氯化钙溶液入口蝶阀(600BF)、壳程冷却水出口闸阀(300GA)，间冷器(4306-COO-001/002/003/004)、电除雾器 (4306-ESP-001/002)、稀酸泵(4306-PSO-001/002)、冷却塔循环槽(4306-TNK-004/005)、逆喷洗涤器循环槽(4306-TNK-006)、填料洗涤塔循环槽(4306-TNK-007)、高位水槽(4306-TNK-003)和水封槽(4306-TNK-009)相对应的阀门，同时，上一级设备均需停车； 当出现紧急事故首先关停风机(4306-FAN-

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№	Stage	Key Points
		001/002/003)，然后再按照停车顺序进行停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，无论时间长短再启动时，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
14	Permits &Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Flue Gas Washing Post 烟气洗涤岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Flue gas washing posts in the chloride plant
CP 烟气洗涤岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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Deputy General Manager 副总经理	Li Xinhua 李新华	Date: 日期 2024.4.24	Signature 签字	
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